



The effect of short formative diagnostic web quizzes with minimal feedback

Olle Bälter^{a,b}, Emma Enström^{a,*}, Bernhard Klingenberg^c

^a School of Computer Science and Communication, KTH Royal Institute of Technology, Stockholm, Sweden

^b School of Education and Communication in Engineering Science, KTH Royal Institute of Technology, Stockholm, Sweden

^c Department of Mathematics and Statistics, Williams College, Williamstown, MA, USA

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ABSTRACT

To help students gauge their understanding of basic concepts and encourage good study habits, we administered short online quizzes that use generic questions in the crucial first few weeks of a course. The purpose of the study was to investigate whether the combination of these web quizzes with generic questions with only binary feedback (right or wrong) would be beneficial for students' learning. We implemented these quizzes in three classes in two different subjects at two different universities, one in Sweden and one in the USA. The students' views on the quizzes' effect on their learning was investigated with surveys and interviews.

Almost all students appreciated having these quizzes and 38% of them changed their view on how much they knew of the material covered in the course. Furthermore, over 20% of the students reported altering their study habits as a consequence, in particular studying harder or earlier.

In conclusion, this study indicated that short quizzes using generic questions with limited correct/incorrect feedback on each question, have positive effects when administered early in courses.

The combination of generic questions and short quizzes could be of value for those contemplating automatic formative assessment, particularly if there is some hesitation with respect to the resources needed for constructing and validating the automatic feedback.

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1. Introduction

During science courses, we often experience students approaching us the last week before the final exam, telling us they did not understand one topic in week 2 and have had difficulties ever since. This is not only a concern with respect to “weak” students; it also happens at top colleges. As teachers, we often suspect certain students are struggling, but we may be wrong and the students may be worried if we approach them asking about their situation. However, we do want to promote continuous learning, and we want to offer support for students who need it. We also want to minimize the time spent, both for students and for teachers, on monitoring progress.

After the first individual assessment, it becomes clear to all how well students are doing. At this point, it may be too late to prevent struggling students from failing (Bergin & Reilly, 2005), if the first individual assessment is a midterm or occurs after a large part of the course has passed. We therefore need to assess students as early as possible with problems suitable to the skill level they should have obtained. We know that a deep approach to learning leads to higher grades (Simon et al., 2006), but also that students' expectation of their own grade (Rountree, Rountree, & Robins, 2002), their comfort level (Wilson & Shrock, 2001) and perception of their own understanding of the module (Bergin & Reilly, 2005) are all important indicators of performance. The initial assessment should therefore be a part of the early identification of struggling students. Passing relevant, but simpler problems (compared to e.g. a midterm) could strengthen their self-confidence and thereby perception of the subject and possibly improve learning in the entire course.

The purpose of this paper is to describe a case study where we used short, online, unsupervised, generic quizzes with limited feedback – correct/incorrect – and this method's effect on students' views of their own knowledge and study habits.

Adoption and integration of Information and Communication Technologies (ICT) in higher education has been slow (Birch & Burnett, 2009). Major barriers for adoption includes lack of a clear institutional strategy and the cost in time and money for developing ICT

* Corresponding author. Tel.: +46 8 790 6540.

E-mail addresses: balter@kth.se (O. Bälter), emmaen@kth.se (E. Enström), bernhard.klingenberg@williams.edu (B. Klingenberg).

resources. Birch and Burnett (2009) suggest remedies including 1) gradually building up ICT resources over time, 2) focus on developing reusable learning objects and, 3) beginning with improving student learning outcomes first.

Highly interactive learning material is one of the great possibilities with ICT, but at the same time it is also costly. On the other hand, modules for assessment are included in all Learning Management Systems (LMS) and can be used with relatively little effort. Previous studies on students' acceptance of computer-based assessment are inconsistent (Fischer & Kopp, 2006; Fluck, Pullen, & Harper, 2009) but actual experience of computer-based assessment seems to increase students' attitudes positively (Deutsch, Herrmann, Frese, & Sandholzer, 2012; Fluck et al., 2009). Despite this, there are strong reservations when using test performance for summative assessment; however optional formative computer-based assessment early in higher education is a "promising possibility" and may be a fruitful implementation strategy (Deutsch et al., 2012).

There is discussion among scholars (Taras, 2005) on the differences between formative and summative assessment and the distinction is not always clear. Taras' view is that formative assessment encompasses summative assessment, but in addition requires feedback which indicates a knowledge gap and also an indication on how to close this gap. Sadler (1989) makes the distinction that formative assessment is used by the learner. Therefore, according to Sadler, summative assessment does not exclude feedback, and the same assessment can be either summative or formative depending on what the student does with it.

For online environments the formative feedback needs to be prompt and offer possibilities to either repeat and/or revise tasks (Gikandi, Davis, & Morrow, 2011). Formative assessment is perceived as useful by students (Cassady & Gridley, 2005). Among the benefits are an increase in learner engagement (Gikandi et al., 2011) and an improvement of results in the final exam (Cassady & Gridley, 2005; Ibabe & Jauregizar, 2009; Wilson, Boyd, Chen, & Jamal, 2011).

Dopper and Sjoer (2004) and Kibble et al. (2011) used formative assessment in the form of quizzes with good results. However, from the Gikandi et al. (2011) overview of formative assessment it is clear that there are many things that can go wrong. Feedback in formative assessment should be clear, timely, ongoing and adequately detailed. Reliability could be threatened unless the assessment offers multiple opportunities to demonstrate students' learning.

While 70% of the students in the Dopper and Sjoer (2004) study found the feedback useful or very useful, the Kibble et al. (2011) study reports a large number of students that did not use the feedback at all.

So, considering the possible positive benefits from formative online assessments, but also the huge investment in developing appropriate feedback for every possible answer, we propose a simple approach that requires less investment. We propose:

1. Taking advantage of the possibilities that come from generic questions using variables and thereby giving students the opportunity to retake the quiz as many times as they like to encourage constructive learning. A similar approach was used by Angus and Watson (2009) who used variables to generate 80 different numerical variations of questions thus making the two allowed attempts at their one hour quizzes different.
2. Giving students the opportunity to retake the quiz as many times as they like to encourage constructive learning (in contrast with the two attempts allowed by both Angus and Watson (2009) and Kibble (2007) and the single attempt by Dopper and Sjoer (2004)).
3. Making these quizzes short so they can be answered in less than approximately 10 min, for those that know the subject, to encourage both taking the test at all and to take the time to do it again (again in contrast with the one hour tests of both Angus and Watson (2009) and Kibble (2007)).
4. Using only binary feedback (correct/incorrect) for each question, in contrast with Dopper and Sjoer (2004) and Angus and Watson (2009) that both used online feedback and Kibble (2007) who used optional face-to-face feedback.
5. Offering quizzes every week for the first 3–4 weeks of a course to encourage good study habits.

With this proposed method we can reduce some of the disadvantages of computer-based assessment, such as the need for more advanced invigilation of students during tests, the consequences of system failures (Cantillon, Irish, & Sales, 2004) and plagiarism (Cantillon et al., 2004; Carter et al., 2003; Daly & Waldron, 2004; Sheard, Dick, Markham, MacDonald, & Walsh, 2002).

The generic features together with the diagnostic nature could also reduce many of the barriers that contribute to test anxiety, see Cassady and Gridley (2005) for an overview.

Here we present results from a feasibility study using generic quizzes in two different university settings.

2. Background

A common method of performing efficient formative assessment is to automate. Computer Aided Assessment (CAA) is often presented as the primary solution for education of the future. Kyllonen (2009) discusses the substance in these claims; there are benefits, but sometimes disadvantages. For some purposes, for instance when an audio or video file helps, CAA provides a more realistic situation than "traditional" assessment, and the 24/7 availability of online formative assessment tasks and the possibility of repeatedly taking a quiz and receiving immediate feedback are among the most important benefits for the learner. However, the share of failing students remains approximately the same (Malmi, Korhonen, & Saikkonen, 2002). The opinions on CAA are split among academics, but it has been suggested that this is mostly due to varying experience with CAA (Carter et al., 2003). Several systems for CAA exist (Benford, Burke, Foxley, Gutteridge, & Zin, 2011; i-assess; Malmi, Karavirta, Korhonen, & Nikander, 2005; Malmi et al., 2002), and there are studies that report no significant difference in examination between online exercising and classroom exercising (Korhonen, Malmi, Myllyselkä, & Scheinin, 2002).

CAA features prominently in programming courses. In such courses, weekly quizzes made in labs can dramatically reduce failure rates (Woit & Mason, 2003). Also, lab exams are better assessors of programming ability than traditional methods such as written exams and programming assignments (Daly & Waldron, 2004). An examination of novice programmers and the SOLO (Structure of the Observed Learning Outcome) taxonomy by Lister, Simon, Thompson, Whalley, and Prasad (2006) however, ends with a recommendation to mix training and assessment of reading and writing tasks; we should therefore not replace written exams entirely with lab exams. Self-assessment has also been used successfully for terminology quizzes as a way to encourage reading lecture material before class (Williams, Bialac, & Liu, 2006).

There are also disadvantages with CAA (Zhang, Zhao, Zhou, & Nunamaker, 2004). If CAA is used on the web, the problem of knowing who is answering the questions and also whether this person is receiving help or not (Carter et al., 2003) becomes difficult. Another difficulty is that almost correct solutions and slightly misconstrued questions, as slips, mistakes and errors, cannot be handled as smoothly as on a written exam or a lab assignment, where the teacher can simply admit the mistake and correct it immediately. In order to improve learning, counter plagiarism and reduce the number of underachieving students (regardless of whether they fail the course or not), we also need to improve assessment.

Various online programming assessment tools exist (Malmi et al., 2005; Roberts & Verbyla, 2003), but unlike the proposal by Roberts and Verbyla (2003), we are only suggesting and evaluating a pedagogical methodology, not a technical system. An argument against combined development and assessment systems suggests the students are not learning to use the tools employed in “real” development. Also, the methodology tested in the present study was not limited to programming, but included a course in applied statistics.

A web-based system for small diagnostic tests would liberate students from coordinating assessments in time and place. Some commercially available systems support this, for instance, *i-assess*.¹ To make cheating by simply copying someone else’s answers a waste of time, systems can use generic questions or Automated Item Generation (AIG) as described by e.g. Kyllonen (2009), Angus and Watson (2009), Kibble et al. (2011). This can be done with different intentions and to different extent. There are usually concerns that varying the questions can substantially alter the difficulty of a test. However, since the tests are for formative purposes and the students can immediately retake the test we believe the variation might be beneficial rather than problematic, as the student in that case would get a wider view on the concepts that the question is about. Our focus was not summative tests, nor adaptive testing, and we are not evaluating questions with respect to psychometric characteristics as e.g. Arendasy and Sommer (2011). Our questions are about the facts and concepts which are central for the course at a certain point in time, and to illustrate these, variation only helps. Since our fields of application are algorithmical, we make use of what Kyllonen (2009) describes as “Slots and Fillers”, i.e. there are obvious places to insert a parameter to vary the question.

For a very simple example, as a first problem in an introductory course in computer science, the following code is provided:

```
x = 17
y = x
x = 42
```

and the question follows: what is the value of *b*? Depending on whether *x* and *y* are primitive or reference variables, the answer will be 17 or 42, respectively. Examining the question we can realize that *x* can be replaced with any valid variable name, as can *y*; and 17 and 42 can be replaced with any numerical value. The foundation of our “Slots and Fillers” questions is that the question is determinable (that is, all answers can be classified as either correct or incorrect) and input dependent (that is, there is input to the question and this input can be varied and affect output).

Web technology makes it possible to give students a small test every day (or before or after a lecture, a lab etc.). A student who fails the test may, thanks to the generic formula, be given a new test immediately in the spirit of constructivism (this idea was proposed by Keller already in 1968 (Herzberg, 2001; Keller, 1968)). Hopefully, this would function as a motivation for students to study, just as other motivators such as narratives, visual programming, robots, Lego (Powers et al., 2006) and visualizations of programs (Naps, Eagan, & Norton, 2000; Rajala, Laakso, Kaila, & Salakoski, 2007).

3. Method

We tested the method in pilot studies and later in three different courses at two different universities and evaluated the results with surveys and interviews, both for the pilots and for the large test version. The participating universities were KTH Royal Institute of Technology in Stockholm, Sweden and Williams College in Williamstown, Massachusetts, USA. KTH is the largest technical university in Sweden with mainly program-based engineering education, while Williams is a top-ranked (Ewalt, 2010) liberal arts college.

At KTH we implemented three diagnostic quizzes for an introductory course in programming methods in Bilda, a KTH version of the Ping-Pong LMS.² This version allows for the usage of JavaScript in the construction and assessment of questions. These quizzes were used in two different classes: one for the 132 students of the program in Industrial Engineering and Management in Fall 2009 and one for the 188 students of the program in Built Environment in Spring 2010.

At Williams College, we conducted a pilot study in an introductory computer science course in Fall 2009 and again in Spring 2010, using the Bilda system at KTH as the technical solution. Unfortunately, due to some logistical difficulties, we abandoned this approach for the Fall 2010, when we implemented four quizzes for a course in applied statistics and data analysis in Glow (Williams College’s version of Moodle).³ This course had 38 enrolled students.

All courses used one quiz per week for at least the first three weeks of these full semester courses. Each quiz included 3–5 questions based on the fundamentals of the previous week’s lectures. After all questions were answered the students were given feedback in form of total points for each question. It was possible to retake the entire quiz immediately, with the questions being slightly changed by the generic features. Students who failed repeatedly (never managed to finish all questions correctly) were contacted via email with an offer of extra assistance. In contrast to Woit and Mason (2003), we chose to offer the tests online for students to take at whatever time they preferred outside labs, so that the time in lab could still be used for interaction with teachers.

Right after the mid-term, students were asked to complete an anonymous online survey distributed via email. The survey comprised of eight questions regarding the students’ views on the quizzes and how these affected their study habits. The survey consisted of multiple choice questions with the possibility to give free text comments on each question. The questions were designed especially for this study and

¹ See references.

² Bilda; URL <http://www.pingpong.se/product.en.html>. Last visited: Oct 4 2011.

³ Moodle; URL <http://moodle.org/>. Last visited: Oct 4 2011.

tested in pilot studies. We used a standard method with one email informing the students of the study, a separate email with the link to the survey and three reminders one week apart for non-respondents.

After the survey, we invited students via email to volunteer for a 30 min telephone, Skype or face-to-face interview concerning their opinion on this way of working and how it affected their study situation.

All students were informed that participating in the survey and interviews of this study was voluntary. The study was approved by the Institutional Review Board at Williams College. The survey was submitted through Bilda.

During the pilot tests, we experimented with different practical arrangements to encourage students' usage of the system. The most important conclusion was to give access to the system outside lab hours, as students otherwise were inclined to skip the quiz in order to finish the lab.

4. Results

At KTH, in all 2009, 102 (77%) of the 132 enrolled students answered the survey. The three quizzes were completed by more than 90% of the students. In the Spring 2010 course, the quizzes first appeared and could be used for bonus points before a certain deadline, and then disappeared. Since this was not appreciated by the students, the quizzes were reintroduced toward the end of the course. Around 50 percent of the students completed each quiz for a bonus, while 70–85% completed each quiz after receiving access to them again before the exam. The survey was completed by 119 (63%) of the 188 students.

At Williams College in Fall 2010, the students had one quiz per week for the first three weeks and the fourth and last two weeks later. Each quiz was completed by between 32 and 35 of the totally 38 enrolled students. The survey was filled out by 29 (76%) students.

For the students, the quizzes were intended to give unbiased, although sparse, feedback about their current knowledge, support choices of what to study more and what to consider appropriately learned. This could possibly affect study habits, self-validation and general stress about one more task to fulfill. Below we present the results with the three student groups separated. As will become clear, the differences between the groups are small; however we believe the similarities are interesting.

As for the students' own perceptions about their current knowledge, see Table 1, the students had various ideas on the possible effect of the quizzes.

When combining the answers from all three courses, a rather large fraction (23%) did not know whether the quizzes affected their evaluation of their own understanding, whereas 38% of them actually adjusted their self evaluation as a result of the quizzes.

Another intention with the generic questions is that the student should be able to repeatedly try one quiz until confident about being able to answer the questions. The option to use the quizzes repeatedly for self-evaluation was appreciated by almost all students (see Table 2). This was evident during the KTH 2010 course, where access to the quizzes was requested by the students after the test period. As mentioned above, the possibility to use the quizzes for studying when the exam was approaching seemingly attracted more students than the bonus points for the exam that could be collected by taking the quizzes on time. However, there were also some issues with missing information regarding the deadlines for the quizzes that might have influenced the early participation negatively.

Some students in the KTH 2009 course were worried that the possibility of actually testing the computer code before answering could be used to pass the quizzes without proper understanding of the material, particularly given the motivation of bonus points. In that situation, it would be unfair to other students who do not cheat, but as pointed out by a student interviewed during the pilot, this is of no real concern since the students who try to cheat will likely cause their own failure later by not learning the material properly. Therefore, if bonus points are used, the level should be high enough to encourage participation in the quizzes, but low enough to not encourage cheating. Also some questions should be posed in a manner that will make simple copy–paste–run insufficient for a correct answer Table 3.

Most participants did not report increased stress as a consequence of taking the quizzes, although about one third of the KTH students did feel a little more stressed. From the comments, it is possible to deduce that many of them found it stressful that the quizzes disappeared after a certain time (usually a week or two), as this made the students feel afraid of unwillingly missing a deadline. The other comments this question received pertain to motivation and study habits.

The question of study habits (see Table 4) received more positive free text comments than the others. More than 20% of the students claimed that their study habits were affected, most of them studying more or earlier, and the comments reveal that they appreciate the opportunity to check their own skills and find gaps or hints on what to repeat.

The answers in Table 4 can be combined with the answers in Table 1. Of those students who indicated that they knew less than they thought after taking the quiz, 40% were motivated to study harder or earlier, but, equally remarkable, another 44% saw no need to change their study habits. Comparing this to the group of students that indicated that taking the quiz confirmed their expectation of how much they know, 15% were motivated to study harder or earlier but 71% did not see a need to change their studying behavior. Finally, for the group of

Table 1
Quizzes' effect on students knowledge. Number and percentage of responses for each option.

Option	KTH 2009	KTH 2010	Williams 2010
Yes, I realized that I knew more than I thought before	21 (21%)	18 (16%)	3 (10%)
Yes, I realized that I knew less than I thought before	23 (23%)	24 (21%)	5 (17%)
No, they confirmed my understanding	36 (35%)	41 (35%)	18 (62%)
No, they did not show what I had learned	n/a	n/a	0
I don't know	22 (22%)	33 (28%)	3 (10%)

The question was: Did the quizzes affect your understanding of how much you knew or how much you had learned about computer science^a?

^a In the Williams survey, "computer science" was replaced with "applied statistics".

Table 2

Importance of generic features. Number and percentage of responses for each option.

Option	KTH 2009	KTH 2010	Williams 2010
Good that I can practice until I understand	97 (95%)	109 (92%)	27 (93%)
It doesn't matter that you can try the tests many times	2 (2%)	8 (6%)	1 (3%)
Allowing several tries is unnecessary	3 (3%)	1 (1%)	1 (3%)

The question was: It is possible to try over again repeatedly on the quizzes, with questions that vary between the tries. How do you feel about that?

students that indicated they knew more than expected after talking the quiz, 26% responded they still would study harder or earlier as a consequence, while 55% said they see no need to change their behavior (and 7% said they would actually study less).

The group of students that reported either more or less knowledge than expected was more than twice as likely to study harder or earlier than the other students. The probability ratio was equal to 2.70, with 95% CI interval equal to 1.66–4.41 and a P -value <0.0001 .

The most common opinion expressed among the optional comments concerning the difficulty of the questions (see Table 5) was that the questions ought to be more difficult and cover more of the course (instead of only the first few weeks).

In general, depending on courses, between 61 and 86% of the students felt that answering the quizzes was an efficient use of their time (see Table 6) often with the motivation that they were easy and did not take much time.

The idea that teachers could offer help to those struggling was appealing to most Swedish students. The students at Williams responded with another pattern, with most answers neutral and a larger fraction negative to the idea, see Table 7. Out of the Swedish students in the two courses, 3 and 14 percent respectively were contacted by their teacher regarding their results. The main reason for this was results came in late – there had been some difficulties logging in and the deadlines were not always clear to all students.

Based on the comments to this question, no one took offense, but the idea “please let me pass or fail the courses without any interventions” existed among the students. Most of the comments were positive on this question as well, but for the majority of the comments, it is not clear whether the commenting person speaks out of experience or hypothesis. One comment states that it should not be necessary to contact people if the bad results are an exception; only frequent failure should result in this course of action as it may be a negative experience to be singled out.

Whether the quizzes yielded bonus points for the exam or not varied between the courses we describe, due to the teachers' preferences. The first and the last course had them, the course in-between (KTH 2010) with the lowest participation did not, but since there are other differences between the courses, this is not necessarily evidence that the lack of bonus points causes lower participation.

4.1. Details from the comment sections of the survey

One student commented that the quizzes set are very similar to programming: you test something and watch the consequences. This is a somewhat worrying answer, as it suggests that programming consists of “trial and error”. On the other hand, the questions are supposed to be phrased so that recognizing a pattern actually constitutes some knowledge of the desired kind, for instance how to call a method of a class – the generalization is the actual goal of the exercise. Trying repeatedly to answer a quiz in order to see this pattern is not necessarily providing a shallower type of knowledge than reading about it in the book and then answering, provided the pattern that is found is a “valid” pattern, one that the teacher agrees exists, and not some misconception. Some practical issues were highlighted, deadlines for bonus points for taking the tests being unclear and the graphical presentation of the questions being ugly, for instance. The students also felt that some questions on the quizzes were asked too soon, i.e. before the subject had been discussed in a lecture; the teachers disagreed. With respect to the unlimited number of tries on each question, and the possible cheating that could occur as a consequence of it, a limit on the allowed number of attempts on each question was suggested.

4.2. Interviews with students

During the Fall 2009 and Spring 2010 four students were interviewed about their experience with the quizzes in order to improve the system and the methodology. During the Fall 2010 five students at Williams College were interviewed. The noteworthy answers are presented below.

4.2.1. Perception of own knowledge

During the pilot, a student at Williams pointed out that the quizzes were very similar to a pre-college experiences of his. At college, students learn to evaluate their progress by themselves, while in high school, there are a lot of feedback stations to pass on the road. All interviewed students during the pilot used the quizzes to confirm that they knew what they should know, and were happy to have the option to use the quizzes. They wanted more quizzes and on another level of difficulty, to be able to diagnose more gaps in their knowledge.

“I like the quizzes, they are quick and easy, It did not take the amount of time the problem set did. It gave you a feel for whether you understood the underlying concepts. It took the pressure off, you could try it again. They did not have a negative impact on your grade.”

Table 3

Quizzes' effect on stress levels. Number and percentage of answers for each option.

Option	KTH 2009	KTH 2010	Williams 2010
Yes, very much	2 (2%)	4 (3%)	0
Yes, a little	31 (30%)	34 (29%)	3 (10%)
No	69 (68%)	79 (68%)	26 (90%)

The question was: Did you find it stressful to do the quizzes?

Table 4
Quizzes' effect on study habits. Number and percentage of responses for each option.

Option	KTH 2009	KTH 2010	Williams 2010
Yes, I studied harder	25 (25%)	11 (9%)	3 (10%)
Yes, I studied less	2 (2%)	5 (4%)	1 (3%)
No, I would have studied just as much/little without the tests	61 (60%)	67 (57%)	21 (72%)
I did not study more, but earlier	n/a	11 (9%)	2 (7%)
I don't know	14 (14%)	23 (20%)	2 (7%)

The question was: Were your study habits affected by the existence of the quizzes or your results on them?

"Yeah, ah. The quizzes helped me understand some things more. You don't understand the whole picture in class. The conceptual questions helped me understand that better. Assurance that I understood the concepts. It kind of ties it up for you in the end. It gives you better confidence."

"Yes. There were some things that the professor K said in class, that I heard but did not understand the importance of, but being asked in the quiz made you think again of what he said. When I did the quiz, I realized why it was important to understand the difference between A and B."

"Yes, during lectures everything is so self-evident, but at the diagnostic quiz it goes wrong. They seem to be easy, but they are tricky."

4.2.2. Generic questions

Although admitting that some questions could be difficult to answer, the students interviewed during the pilot seemed to not use the quizzes for practicing, but for identifying what to repeat and study more on. Once directed toward some of the course content, the student would study that better and often not try the quiz again. Since they either did not try quizzes several times at all, or just not several times in a row, the Williams students in the pilot study hardly noticed that questions changed. This was different during the real test, when the experiment was better aligned with the course.

"You can retake as many times as you want. Some students might not take it seriously. <omitted> Limiting to ten [tries] would be good."

"Easily accessible, easy to take. Number changing between tries, do not change that."

4.2.3. Perceived stress

None of the interviewees found the quizzes stressful:

"It is there! The instant feedback, you can do it multiple times. Less stressful that you do not have to get it right the first time."

"The option of not doing the quiz is important, no pressure to get it right, instant feedback important advantage. Serves a different purpose than homework."

4.2.4. Effect on study habits

No one interviewed during the pilot claimed that his or her study habits had changed because of the results on the quizzes, but they meant that this of course would have been the case, had their results been worse.

"It is really great that it is not mandatory. We discussed these questions in my [study] group and then you understand that you have thought differently and then you get help from others who are a little better."

4.2.5. Teacher contact

During the pilot study, students at Williams felt that the perceived possibility of a teacher contacting them regarding their results on the quizzes was both unlikely (the Williams' culture was that the students asked for help if they needed it), and not perceived as an invasion of their privacy, given the professors would already know how their students did on the course assignments and actually would assign everyone a grade. However, students were not completely comfortable with being contacted by teachers if they failed the quizzes: *"That would not be helpful, better to encourage the student to contact the teacher. Risk the students overstudy for it, and make it a bigger assignment than it should be. TAs contacting the students would be very good. Write up the solutions to the problems and send them to the students. Involving the professor makes it too formal. Almost like a test. The students believe in the subjectivity of the grades."*

An automatic message encouraging the student to contact the teacher could be an alternative:

"Uncomfortable, that would stress me out more. If the professor looked at my actual grades. But, that would mean I would put more effort into it. The professor would have to approach me in a non-threatening way. A nice conversation would be comfortable. Having TAs contact you would be better. More so than the professor."

"It is a good thing. That means the student needs help, but the students may have problems matching the professors office hours, the student may think they will solve it [their lack of knowledge] with time, or they may be too shy to contact the professor."

"Good idea [to contact the failing students]. Some students are hesitating to contact the professor. As a tutor I like that."

"You don't want to be the one doing poorly. You do not want the teacher to know you are struggling. Maybe have the TAs contact the students? Students are surprised to know that teachers love it when they come to office hours. Difficult line to walk between a voluntary quiz and an assessment if the teacher contacts them. TAs could work as a middle man."

Table 5
Questions' level of difficulty. Number and percentage of answers for each option.

Option	KTH 2009	KTH 2010	Williams 2010
Yes	88 (86%)	79 (68%)	25 (89%)
No	7 (7%)	11 (9%)	2 (7%)
No opinion	7 (7%)	27 (23%)	1 (4%)

The question was: Did you consider the difficulty of the questions being suitable for everyone following the class to be able to answer them with reasonable effort?

Table 6
Quizzes' efficiency. Number and percentage of answers for each option.

Option	KTH 2009	KTH 2010	Williams 2010
Yes	73 (72%)	72 (62%)	25 (86%)
No	9 (9%)	15 (13%)	1 (3%)
I don't know	20 (20%)	30 (26%)	3 (10%)

The question was: Do you think that the quizzes were an efficient use of your time?

"It would have been nice with an email before the teacher contacted me. Now I was a little put out (one of the few students that was contacted directly)."

Both KTH and Williams students seemed a bit negative to the idea of being contacted by a teacher, despite the answers on the survey. At KTH, students' attitudes were close to "please let me fail the course undisturbed" whereas the Williams pilot interviews explained that, at Williams, it would be a very severe thing if a teacher would approach someone with any message regarding how courses went. Teachers tried to silently notice if someone was struggling and pay the struggling students more attention without mentioning this to them, and students were expected to contact their teachers if not everything was going great.

4.2.6. Others

In the interviews we also asked the students some questions that were not based on the results of the surveys and the respondents were also encouraged to give open views on the idea. The first interview question was: "Can you compare these web-based quizzes with other forms of assignments, such as homework, mid-terms, etc? Pros and cons?"

"I put less effort into the quizzes, when it comes to homework I think it through. The quizzes I do as quickly as I can. Definitely less effort compared to other assignments. I do it quickly, if I get it wrong – I can do stupid errors on the homework, but here I can always go back and redo the problems. Extra credit boost my homework score, it actually helps me to understand what I know and what I don't know. I get a deeper understanding of it."

"Pro: Since it is on the computer, they are fairly easy to do. Con: I just click, and not just think about it. I put less effort into my answer. Pro: it saves time. Instant feedback."

Another question was: "How would you react to an additional question that was voluntary, but more difficult?" inspired by suggestions from previous interviews and surveys.

"As an extra credit thing, may encourage the students to do it. Without the extra credits, would make it more formal."

"I think I would like that a lot. More extra credits are always nice. More problems at the same level as mid-term or finals level. Would not do it without the extra credit, unless I had the time. It would be an improvement."

"Probably helpful with an optional question and to do it as many times you need."

"It would be good to have an extra difficult problem. It is just fun to do! No extra credits are necessary."

Another question inspired by previous comments from the participants was: "How would you react on reminders as the deadline approaches?"

"The first couple of times we were uncertain of the closing time. And sometimes I forget, and then it would be great to have a reminder. Would be great if the quiz did not overlap with the problem set to encourage continuous work."

"Reminders are nice. One or two. 1 hour before. 2–3 hours maybe."

"One email reminder would be great. 24 h warning."

"Reminders are great. Preferably 2 days in advance."

If reminders are to be used it is apparent that the preferred time varies quite a lot between students.

When asked, "How would you change the system if you could?" few suggestions came up, but one student proposed two ideas: *"If you kind of gave a bonus for getting it right the first time. Limiting the number of tries to ten would also be fine."*

Finally, we asked "What do you consider the best feature of the system?" and the opposite question. Nothing noteworthy came out of the last one, but the positive question gave these answers: *"Acts as another reminder. Constant reinforcement of what you are doing."*

"It is quick, you can answer again, they take less time, you can practice on concepts, problems set just a lot longer."

"Do it, see the question, the input and the answer. On the computational ones no more feedback is necessary. For the conceptual ones, it might be difficult to do without giving away the answer. After the quiz closes, send out answers with commentary. Make me figure it out myself helps me learn it better."

4.3. Interviews with teachers

The five teachers responsible for the courses used for the pilot and the study were interviewed after the quizzes ended for each course about their experience with the quizzes in order to improve the system and the methodology. Being colleagues with the authors, their answers may be biased, but all of them expressed interest in continuing using the web-based quizzes (and one of them has already used it without the authors' assistance).

Table 7
Teachers contacting failing students. Number and percentage of the respondents for each option.

Option	KTH 2009	KTH 2010	Williams 2010
Good	72 (71%)	69 (59%)	9 (31%)
No opinion	27 (27%)	39 (33%)	16 (55%)
Not good	3 (3%)	9 (8%)	4 (14%)

The question was: How do you feel about teachers contacting students that fail the quizzes?

One of the main expected benefits for the teacher was to identify students in need of help and offer assistance early to those students. However, although all failing students at KTH were offered assistance, only one student accepted it. The remaining students used a large variety of explanations why they had failed the quizzes (most common seem to be: forgot or did not have the time). One of the teachers mentions the possibility that it may actually result in fewer questions from the students as there is usually a large group of students who ask questions not in order to gain new knowledge, but to confirm that they have understood. With a 100% score on three quizzes this need for reassurance has totally disappeared. According to these teachers, the quiz system seems to deliver what it promised: uncertain students gain either self-confidence or insight through the quizzes.

5. Discussion and conclusions

We have presented a study of students at two universities who were offered a generic web-based diagnostic test each week for the starting weeks of three courses, two in programming the third in applied statistics. The main intentions with the tests were to encourage students to study early on in the courses, achieve a better view on their own knowledge and assist teachers in identifying students in need of extra support. Ultimately, we hope that this will lead to better learning and study habits among our students. The last two effects are beyond the scope of this case study, but the surveys and interviews all indicate that the system used was appreciated by the students and had the intended effect on many (or most, depending on the question). Improved learning from formative assessment, in form of improved results on the final exam has been shown in studies (Cassady & Gridley, 2005; Ibabe & Jauregizar, 2009; Wilson et al., 2011).

The teachers responsible for the courses did mention that their perception was that the students' results had improved compared to previous classes. Proving that this was indeed so, and was so due to the quizzes, would both be most interesting, but also a monumental task that was never intended to be included in this study. Still we believe the results from the survey and interviews clearly supports our conclusions.

In summary, it appears that the online quizzes do succeed in guiding the study behavior in the right direction.

Depending on the course, between 27 and 43% of the students changed their perception of their own knowledge in either direction (they knew more or less than they thought) and both directions must be considered as positive. Students realizing they know more than they thought will strengthen their self-esteem (Bergin & Reilly, 2005; Rountree et al., 2002; Wilson & Shrock, 2001). Students realizing their lack of knowledge will be aware of that (and hopefully study more).

When it comes to changing study habits more than 20% answered that they had changed them. Of these, almost all studied more or earlier, but a handful of students claimed the quizzes made them study less. However, this may not necessarily be negative, if these are students who need to spend more time on other subjects.

Whether these numbers are impressive or not is up to each teacher to decide, but we should bear in mind that all students involved in this study came from highly ranked universities with a selective admission process. Also bear in mind that a change of study habits is no goal per se – it depends on the present study habits. It is no surprise that a majority of students already had a good understanding of their own knowledge and saw no need to change their study habits. The lowest numbers for these changes also came from Williams College with the most selective admission process. In addition, if we had not used the quizzes, no students would have changed their views on their knowledge nor their study habits.

The generic feature that allowed students to redo the tests was appreciated by more than 92% in all three student groups.

The only question for which there really is a noteworthy difference between KTH and Williams College is the question on whether the teacher should contact students failing the quizzes. This seems to be a more sensitive question at Williams, maybe due to the more competitive grading situation (grade inflation is not an issue at university level in Sweden).

A possible feature that could be added to the system is to limit the number of tries on the generic questions to ten in order to encourage students to think before answering. This feature was already discussed at the planning phase of this project, but was disregarded. However, since none of the participating students needed more than five attempts on any quiz, a limit of for example ten attempts could be investigated. Also, adding more challenging questions outside the quizzes could inspire some high achievers to further studies.

As a note from the authors implementing questions: They were implemented in two different LMS's, the Ping-Pong system used at KTH and Moodle used at Williams College. There are of course major differences between the systems, but with regard to this study the important difference is the possibility of posing generic questions. In Moodle this can be achieved with wildcards, which is sufficient for simple questions. However, for questions involving data sets such as vectors and matrices the possibility of using general programming language as in KTH's extension of Ping-Pong was invaluable. We therefore strongly recommend anyone following our suggestion to make sure a general purpose programming environment is available in the LMS. Similar issues have been reported by Angus and Watson (2009) in their use of the software Respondus: simple variables were supported, but matrices needed work-arounds.

Due to the reduction of questions between classes mentioned by one of the interviewed teachers, we will continue this work by adding a generic quiz to the beginning of a communication course given to computer science students. The course divides these students into two groups: those that understand its importance and those who don't. We hope that this quiz will make the latter group understand that they have things to learn from the course and therefore change their attitudes.

There are several studies that report positive results from formative assessment from systems that give feedback based on the students' answers (Cassady & Gridley, 2005; Dopper & Sjoer, 2004; Gikandi et al., 2011; Ibabe & Jauregizar, 2009; Kibble et al., 2011; Wilson et al., 2011). In this study, we used generic possibilities in combination with short quizzes, administered early with limited feedback (correct/incorrect) on each question and achieved similar results. This combination of generic questions and short quizzes could be of value for those contemplating automatic formative assessment, but who may be hesitating due to the resources needed for constructing and validating the automatic feedback. These results might reduce the barriers for ICT adoption in higher education reported by Birch and Burnett (2009).

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